

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Apple, OR -- station KDLS, America/Los_Angeles
Committed pair OSM 706759844 -> 1276373853
Apple / Apple
Facade gap 84.9 m between the two halls
Weather record 42,451 real hourly records from KDLS
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4502 C at 310 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-05-21 (knife_edge)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. The reactive incumbent took 9 hours -- 3 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	12.598	18.0	12.637	1.363
01	mechanical	yes	switch budget	12.345	18.0	12.108	1.333
02	mechanical	yes	switch budget	12.356	18.0	12.052	1.285
03	mechanical	yes	switch budget	12.619	18.0	12.052	1.154
04	mechanical	yes	switch budget	12.034	18.0	11.518	1.400

05	mechanical	yes	switch budget	12.034	18.0	11.518	1.298
06	mechanical	yes	switch budget	12.418	18.0	11.485	1.271
07	mechanical	yes	switch budget	13.695	18.0	12.569	1.401
08	mechanical	yes	switch budget	15.967	18.0	14.822	1.957
09	mechanical	yes	switch budget	17.328	18.0	15.909	2.114
10	mechanical	no	dry-bulb	19.270	18.0	17.569	2.034
11	mechanical	no	dry-bulb	20.123	18.0	17.019	2.033
12	mechanical	no	dry-bulb	21.741	18.0	19.239	2.125
13	mechanical	no	dry-bulb	22.077	18.0	19.196	2.153
14	mechanical	no	dry-bulb	21.759	18.0	20.349	2.081
15	mechanical	no	dry-bulb	20.859	18.0	18.086	1.774
16	mechanical	no	dry-bulb	19.442	18.0	16.416	1.619
17	mechanical	no	dry-bulb	19.932	18.0	17.628	1.541
18	FREE	yes	-	17.350	18.0	16.485	1.241
19	FREE	yes	-	16.567	18.0	16.485	1.597
20	FREE	yes	-	17.136	18.0	16.518	1.743
21	FREE	yes	-	16.461	18.0	15.935	1.831
22	FREE	yes	-	16.310	18.0	15.359	1.603
23	FREE	yes	-	16.303	18.0	15.359	1.423

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.598 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.345 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.356 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.619 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.034 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a

thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.034 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.418 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.695 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.967 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.327 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.270 C against a 18.0 C limit, so it fails by 1.270 C. It would take a limit 1.270 C higher, or a bound 1.270 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.123 C against a 18.0 C limit, so it fails by 2.123 C. It would take a limit 2.123 C higher, or a bound 2.123 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.741 C against a 18.0 C limit, so it fails by 3.741 C. It would take a limit 3.741 C higher, or a bound 3.741 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.077 C against a 18.0 C limit, so it fails by 4.077 C. It would take a limit 4.077 C higher, or a bound 4.077 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.759 C against a 18.0 C limit, so it

fails by 3.759 C. It would take a limit 3.759 C higher, or a bound 3.759 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.859 C against a 18.0 C limit, so it fails by 2.859 C. It would take a limit 2.859 C higher, or a bound 2.859 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.441 C against a 18.0 C limit, so it fails by 1.441 C. It would take a limit 1.441 C higher, or a bound 1.441 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.932 C against a 18.0 C limit, so it fails by 1.932 C. It would take a limit 1.932 C higher, or a bound 1.932 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.350 C, which is 0.650 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.241 C, and the margin is measured, not chosen: 0.880 C of group-conditional forecast error for this hour of day, plus 0.3606 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.567 C, which is 1.433 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.597 C, and the margin is measured, not chosen: 1.229 C of group-conditional forecast error for this hour of day, plus 0.3677 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.136 C, which is 0.864 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.743 C, and the margin is measured, not chosen: 1.347 C of group-conditional forecast error for this hour of day, plus 0.3959 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.461 C, which is 1.539 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.831 C, and the margin is measured, not chosen: 1.486 C of group-conditional forecast error for this hour of day, plus 0.3450 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.310 C, which is 1.690 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.603 C, and the margin is measured, not chosen: 1.222 C of group-conditional forecast error for this hour of day, plus 0.3807 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.303 C, which is 1.697 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.423 C, and the margin is measured, not chosen: 1.051 C of group-conditional forecast error for this hour of day, plus 0.3728 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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